

Software Requirements Specification (SRS)

Project Title: Smart Digital Microscope

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the **Smart Digital Microscope** project — a web-based intelligent microscopy system capable of capturing, detecting, and classifying white blood cells (WBCs) from microscope samples using artificial intelligence.

The system automates WBC counting and classification into 11 categories and generates medical-style analytical reports, assisting medical and educational laboratories in performing efficient and accurate microscopic analysis.

1.2 Scope

The **Smart Digital Microscope** integrates hardware (Raspberry Pi with a camera module) and AI software to:

- Capture microscopic images automatically.
- Detect white blood cells using **YOLOv8** detection model.
- Classify each detected cell into one of 11 classes using **EfficientNetB3** classifier.
- Stop capturing automatically after reaching 100 WBCs or a user-defined count.
- Generate analytical results including total WBC count, percentages, absolute values, and line chart visualizations.
- Store reports in a database and allow users to view or download them as PDFs.
- Provide a secure login system and dashboard for managing tests and user data.

The system will run primarily on **Raspberry Pi Zero**, with a web-based user interface built using **Python Flask**, accessible on multiple devices within a local network.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
WBC	White Blood Cell
AI	Artificial Intelligence
CNN	Convolutional Neural Network
YOLOv8	You Only Look Once, version 8 (object detection model)
EfficientNetB3	A CNN architecture for image classification
Flask	Python-based web framework
T.WBC	Total White Blood Cell Count
GUI	Graphical User Interface

1.4 References

- Kaggle Dataset: *Blood Cells Dataset (11 Classes – 26,534 images)*
- Laboratory reference document: *Severe Leukocytosis PDF (provided)*
- Internal design notes and user responses.

1.5 Overview

The remainder of this document describes:

- System features and behavior.
- Hardware and software interfaces.
- Functional and non-functional requirements.
- Constraints and assumptions.

2. Overall Description

2.1 Product Perspective

The system acts as an intelligent extension to a traditional optical microscope. It captures digital images through a **Raspberry Pi Camera Module**, processes them locally using embedded AI models, and displays the classified results through a Flask-based web dashboard.

The system integrates image processing, AI inference, data storage, and report visualization in one automated workflow.

2.2 Product Functions

The main functions include:

1. Automatic or manual image capture from a microscope camera.
2. Real-time detection and counting of WBCs using YOLOv8.
3. Classification of each cell into one of 11 categories using EfficientNetB3.
4. Automatic stopping of capture when reaching 100 WBCs.
5. Calculation of:
 - Total WBC count (T.WBC).
 - Percentage of each WBC class.
 - Absolute values (#) in $\times 10^9$ cells/L.
6. Generation of line chart and numerical report.
7. Saving and retrieving reports in a database.
8. Exporting results as PDF.
9. Multi-user system with secure login.
10. Dashboard with settings for capture mode and AI sensitivity.

2.3 User Classes and Characteristics

User Type	Description
Laboratory Technician	Operates the microscope and runs tests.
Medical Student	Uses system for learning and training purposes.
Supervisor / Doctor	Reviews reports and manages user access.
System Administrator	Manages database and user authentication.

2.4 Operating Environment

- **Hardware:** Raspberry Pi Zero, Raspberry Pi Camera Module.
- **Software:** Python 3.10, Flask, TensorFlow, PyTorch, YOLOv8, OpenCV, SQLite.
- **Platform:** Raspberry Pi OS / Linux environment.
- **Network:** Local Wi-Fi network for accessing web interface.
- **Input Device:** Microscope with connected camera (USB or GPIO).
- **Output:** Web interface (Desktop or Mobile browser).

2.5 Design and Implementation Constraints

- Limited GPU performance on Raspberry Pi Zero → lightweight models required.
- Dataset limited to Kaggle source (11 classes).
- System must operate fully offline (no cloud dependency).
- English-only interface.

2.6 Assumptions and Dependencies

- Optical microscope used is compatible with Raspberry Pi camera mount.
- Adequate lighting and focus provided for accurate image capture.
- Network connectivity available for accessing Flask dashboard.
- Power supply is stable for Raspberry Pi.

3. Specific Requirements

3.1 Functional Requirements

FR-1: Image Acquisition

- System shall capture images automatically or manually via camera control.
- Camera shall support zoom, focus, and snapshot control through the UI.
- Capture process stops when total detected WBCs ≥ 100 (default) or user-defined limit.

FR-2: Preprocessing

- Captured images shall undergo color correction, contrast normalization, and resizing before AI inference.
- Optionally convert to grayscale for model input.

FR-3: WBC Detection

- System shall use YOLOv8 for WBC localization.
- Detected bounding boxes shall be cropped and sent to the classifier.

FR-4: WBC Classification

- Each cropped image shall be classified by EfficientNetB3 into one of 11 WBC categories:
- Basophil, Banded Neutrophil, Eosinophil, Erythroblast,
- Lymphocyte, Meta-myelocyte, Monocyte, Myelocyte,
- Platelet, Pro-myelocyte, Segmented Neutrophil

FR-5: Counting and Calculation

- System shall count total WBCs detected.
- System shall calculate:
 - Percentage of each class.
 - Absolute count (#) per $\times 10^9$ cells/L.
 - Total WBC (T.WBC) value (e.g., 7.78).

FR-6: Report Generation

- Generate detailed report containing:
 - Sample image.
 - WBC types (% and #).
 - Interpretations from medical reference file (e.g., “Moderate Leukocytosis”).
 - Line chart visualization.
- User may view the report or export it as PDF.
- Reports shall be saved in database for future reference.

FR-7: User Management

- Secure login system with roles (Technician, Doctor, Admin).
- Each user’s reports stored under their account.

FR-8: Settings and Dashboard

- Provide dashboard for changing:
 - Capture mode (single image / video stream).
 - Model sensitivity and zoom level.
 - AI parameters and report templates.

FR-9: Multi-device Access

- System shall be accessible via multiple devices on the same network (desktop or mobile browser).

3.2 Non-Functional Requirements

ID	Category	Description
NFR-1	Performance	System shall classify at least 10 frames per second on Raspberry Pi Zero.
NFR-2	Accuracy	Minimum classification accuracy $\geq 90\%$.
NFR-3	Usability	Intuitive English web interface with clear charts and status indicators.
NFR-4	Security	Password-protected user accounts; no public access without login.
NFR-5	Scalability	AI model structure shall allow adding more WBC classes in the future.
NFR-6	Portability	Compatible with Raspberry Pi OS and Linux systems.
NFR-7	Reliability	Continuous operation for 2+ hours without crash.
NFR-8	Maintainability	Modular architecture separating AI, Camera, and UI components.
NFR-9	Availability	System uptime $\geq 95\%$ in lab environment.

4. External Interface Requirements

Interface	Description
Hardware Interface	Raspberry Pi Camera connected via GPIO; microscope optical output aligned with camera lens.
Software Interface	Flask web app, TensorFlow & PyTorch models, SQLite database.
User Interface	Web dashboard for capture, analysis, reports, and user management.
Communication Interface	HTTP (localhost or LAN IP) for browser access.

5. System Architecture Overview

Components:

1. **Camera Module** – Captures microscopic images.
2. **YOLOv8 Detection Engine** – Detects WBCs in images.
3. **EfficientNetB3 Classifier** – Classifies each detected WBC type.
4. **Flask Web Server** – Provides UI, API endpoints, and handles user sessions.
5. **SQLite Database** – Stores users, results, and reports.
6. **PDF Generator** – Converts analysis results into downloadable reports.

6. Future Enhancements

- Expand classification from 11 to additional WBC types.
- Support dual-language (Arabic/English) interface.
- Integrate advanced camera autofocus algorithms.
- Mobile application extension for remote access.
- Integration with hospital Laboratory Information Systems (LIS).

7. Appendices

- Dataset Reference: Kaggle "Blood Cells Dataset – 11 Classes".
- Medical Reference: "Severe Leukocytosis" classification ranges.
- Hardware: Raspberry Pi Zero, Camera Module, Optical Microscope, Power Supply.
- Development Tools: Python, TensorFlow, Flask, OpenCV, YOLOv8, Chart.js.

✅ **Document Owner:** 4F AI Team

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📘 **Project:** Smart Digital Microscope